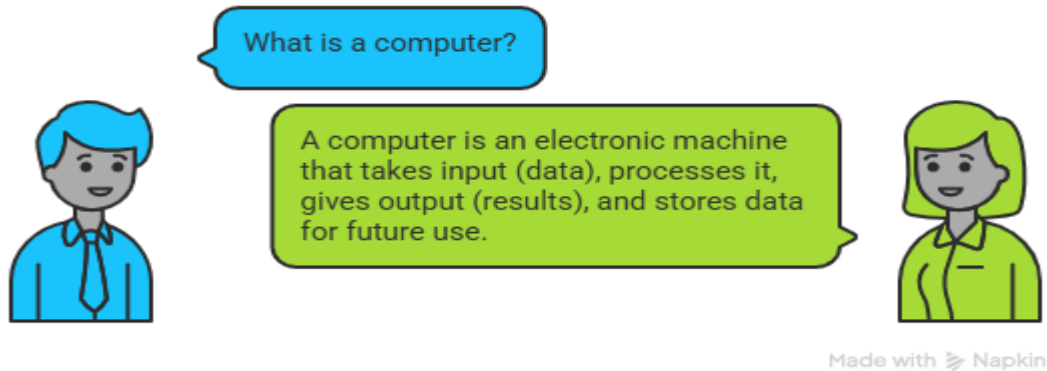


Computer Fundamentals for Beginners

1.What is Computer?

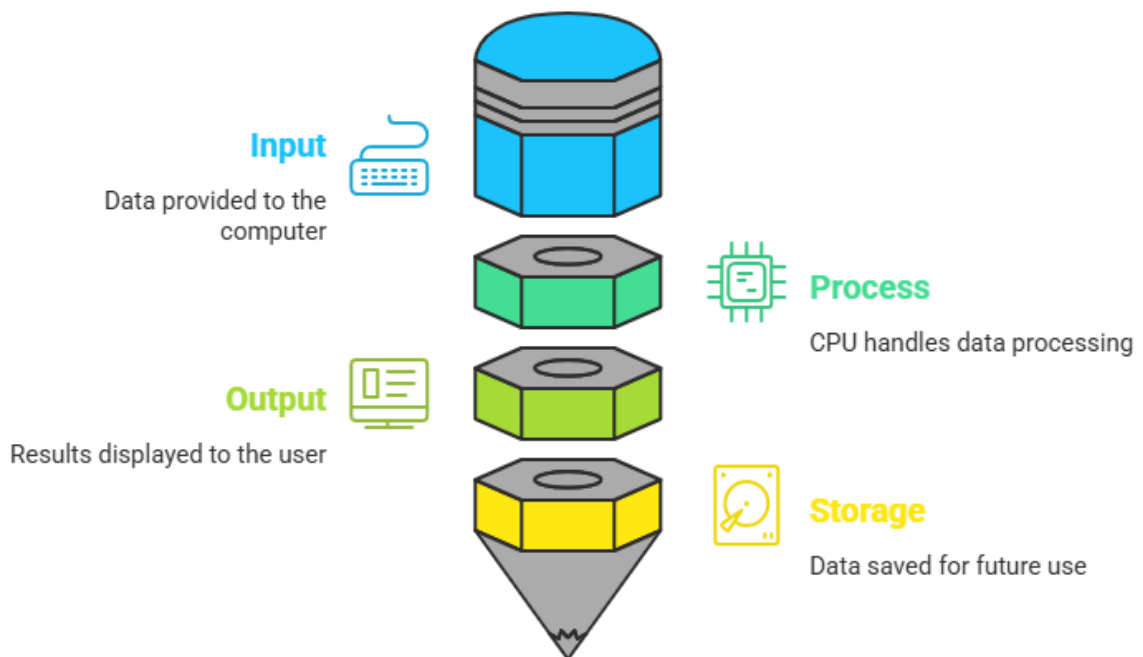
A computer is an electronic machine that takes input(data), processes it gives output(results), and it stores data for future use.

What is a Computer?



2.How a Computer Works(Basic Cycle)

The Computer's Basic Cycle

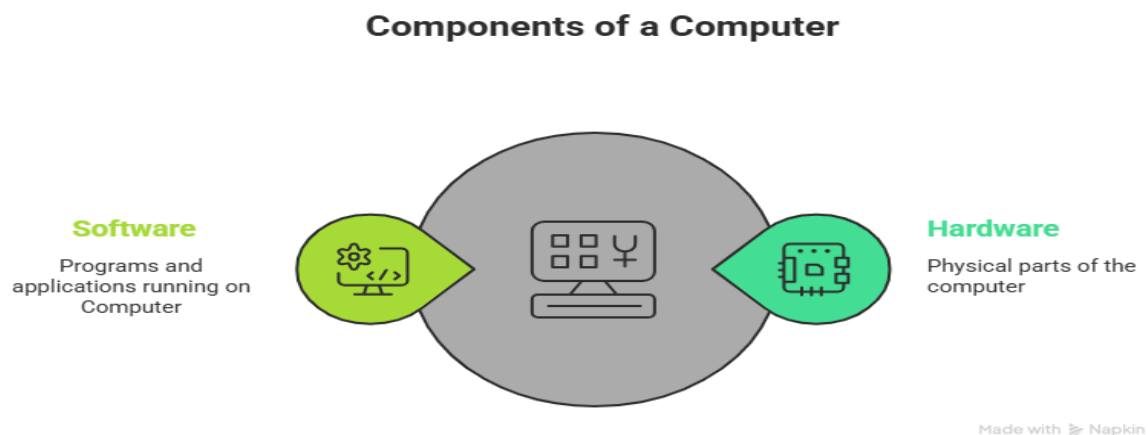


Every computer works in 4 Steps:

- 1.Input: Data given to computer.
- 2.Process: Computer processes data using CPU(Central Processing Unit).
- 3.Output: Result shown to user.
- 4.Storage: it is used to save the data for future use.

3.Main Components of a Computer

The main components of a computer is Hardware and Software components.



1. Hardware(Physical Parts): Things you can touch like Monitor, Keyboard, Mouse and CPU is called Hardware components.

2. Software(Programs): Programs that run on computer is called Software components.

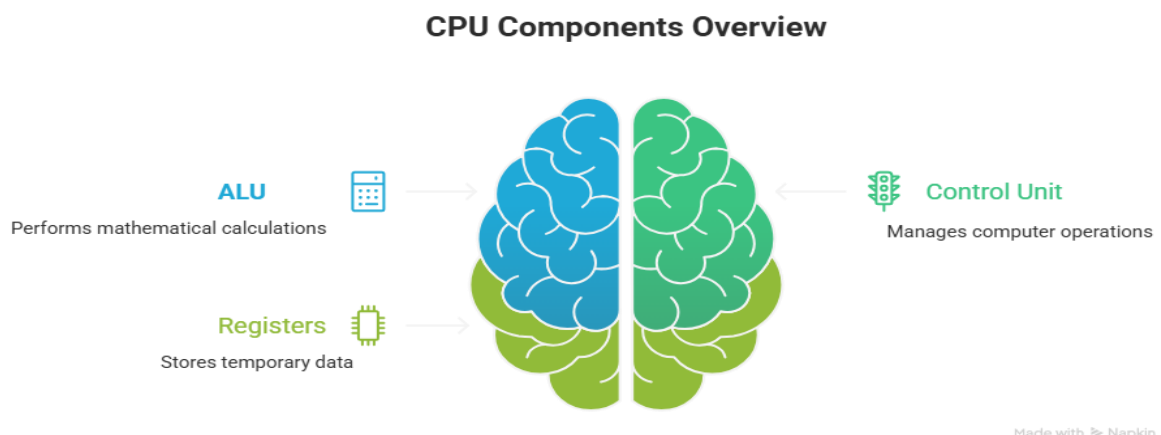
*Without software, hardware cannot work.

Operating System (Windows, Linux, and MacOS)

Applications (Chrome, MS Word, and Excel)

4.CPU(Central Processing Unit)

CPU is the brain of the computer and it performs all calculations and decisions. It has 3 parts:



1. ALU (Arithmetic Logic Unit): Performs math calculations.

2. Control Unit(CU): it is used to control all the operations inside the computer.

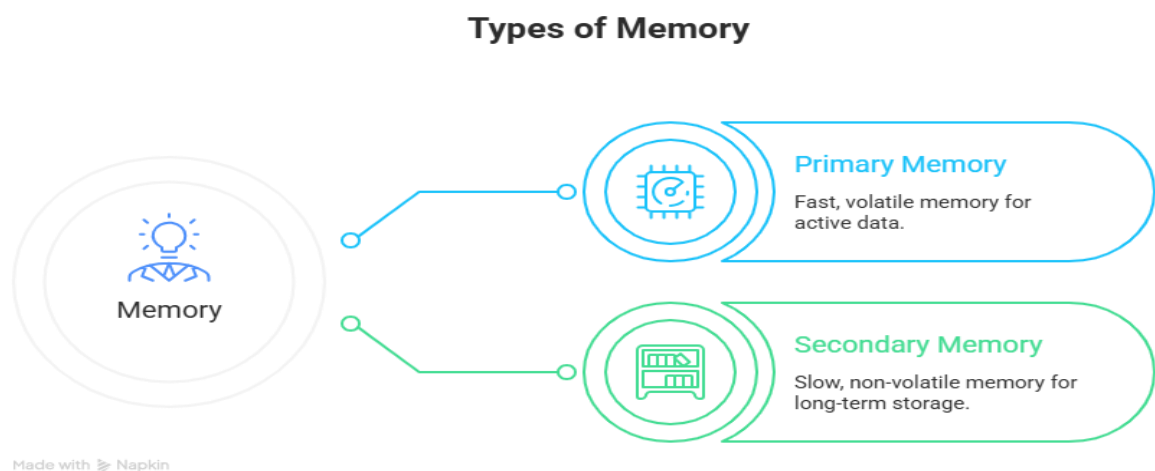
3. Registers: register is small temporary memory.

5. Memory

Memory is the storage area where data and instructions are stored.

Memory are 2 types:

- ❖ Primary Memory
- ❖ Secondary Memory



1. Primary Memory(Main Memory): Primary memory is the main memory of a computer that stores data and instructions temporarily while the computer is running. It is directly accessed by the CPU and very fast

Why we use it:

We use primary memory to store the data and programs that are currently being used so the computer can process them quickly.

✚ **RAM(Random Access Memory):** Temporary memory used while computer is running.[Data lost when power OFF]

Why we use it:

To store current work

For fast access

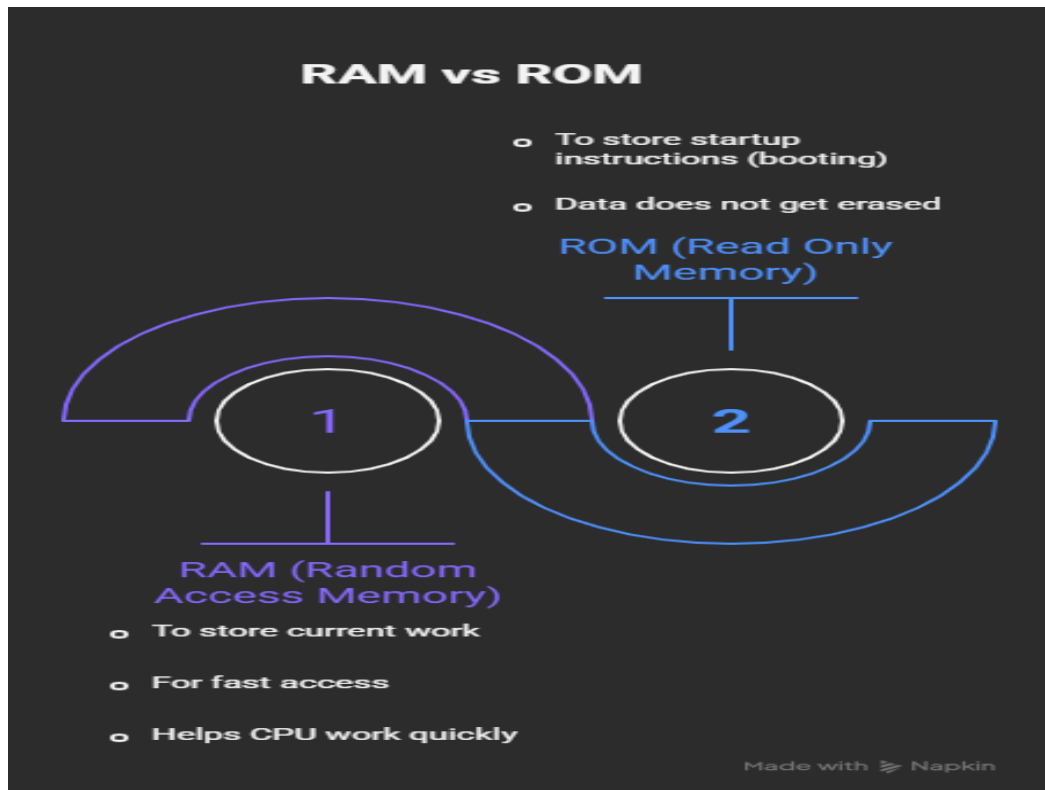
Helps CPU work quickly

✚ **ROM(Read Only Memory):** permanent memory that stores important instructions

Why we use it:

To store startup instructions(booting)

Data does not get erased.



2.Secondary Memory: Permanent storage used to store data for long period of time.

Why we use it:

We use secondary memory to store files, software, and data permanently, even when the computer is turned off.

✚ **HDD(Hard Disk):**it is a secondary storage device that stores data permanently using rotating disk(magnetic storage).

Why we use it:

To store large amounts of data(files, videos, software)

It is cheaper compared to SSD

Used in computers for long-term storage

✚ **SSD(Solid State Drive):** an SSD is a storage device that stores data using flash memory(no moving parts).

Why we use it:

Very fast performance

Faster booting and loading

More durable.

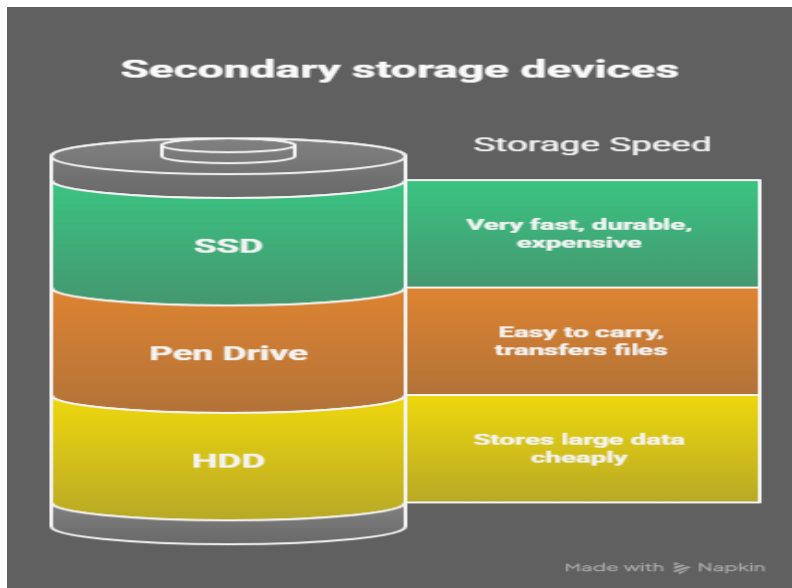
✚ **Pen Drive(USB Flash Drive):**it is small portable storage device that uses flash memory and connects through a USB port

Why we use it:

Easy to carry data anywhere.

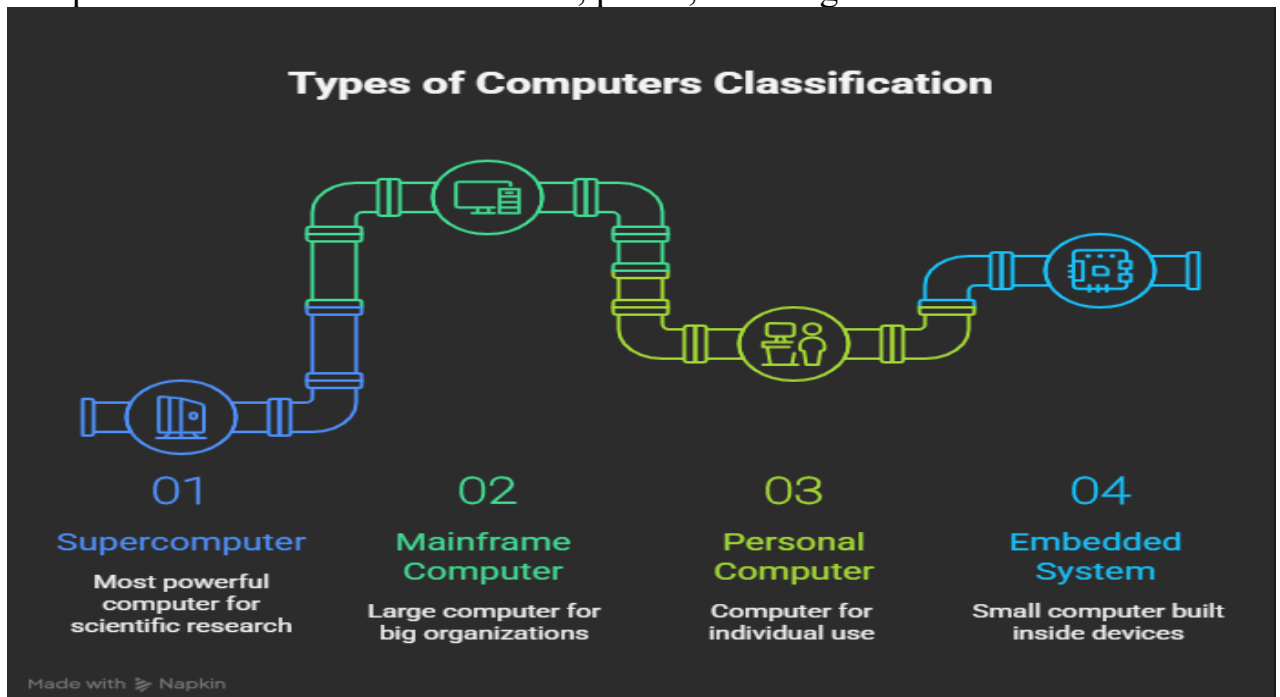
Used for files transfer between devices.

Plug and play(no installation needed)



6.Types of Computers

computers are classified based on size, power, and usage.



Types:

- 1. Supercomputer:** it is a most powerful type of computer that can process huge amounts of data very quickly.

Why we use it:

It is used for scientific research, weather forecasting, and space studies.

- 2. Mainframe Computer:** it is a large and powerful computer used by big organizations to handle large data.

Why we use it:

It is used in banks, airlines, and companies for processing large transactions.

3. Personal Computer(PC): it is a computer designed for individual use, such as desktops and laptops.

Why we use it:

It is used for study, office work, internet browsing, and entertainment.

4. Embedded System: it is small computer built inside a device to perform a specific function.

Why we use it

It is used in machines like washing machines, cars and microwaves to control operations.

7. Basic Computer Terms

- **Bit:** it is the smallest unit of data in a computer, and it can be either 0 or 1.
- **Byte:** a byte is a group of 8 bits used to represent a single character like a letter or number. (8 bits = 1 byte)

File: a file is a collection of related data stored on a computer.

- **Program:** a program is a set of instructions given to a computer to perform a specific tasks.